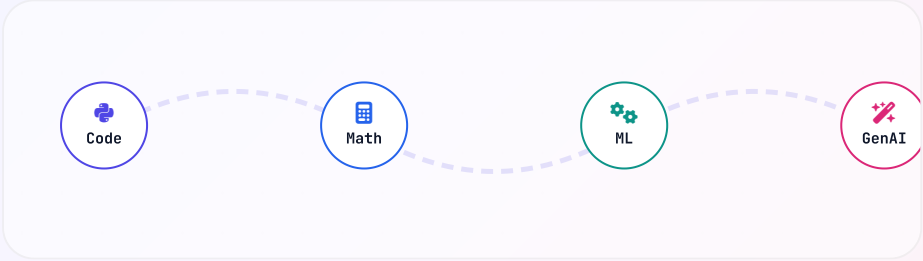
ROADMAP

CAREER BREAKTHROUGH

You Don't Need a Degree to Break Into AI.

You Need a Roadmap. Here's the exact path. Step by step. Every resource is free. No experience needed.



Swipe to start →

MYTH BUSTER

First, Let's Kill the Myth



> *"I need a Computer Science degree to learn AI."*

Stop. That's not true anymore. The field has never been more accessible. Some of the best AI practitioners today started as:

- → Designers: curious about data
- → Teachers: to understand ChatGPT
- → Doctors: exploring medical AI
- → Beginners: zero coding background

NO CREDENTIALS

Free courses from MIT, Stanford, Harvard, and Google exist right now. You just need to know where to look — and in what order.

What separates people who break into AI from those who don't isn't talent. It isn't a degree. It's having a clear path and the discipline to follow it. This is that path.

Before You Start: The Right Mindset



> Most beginners make the same mistake. They try to learn everything at once. They jump between courses. They watch tutorials without building anything. They wait until they "feel ready."

Here's the truth: You will never feel ready. You just have to start.

A FEW RULES BEFORE YOU BEGIN

- **One resource at a time.** Finish it. Then move.
- **Build something every week.** Even if it's tiny.
- **Google is not cheating.** It's a skill.
- **Confusion is not failure.** It's the process.
- **Consistency beats intensity.** 1 hour daily beats 8 hours on Sunday.

Now let's build. 🚀

STEP 1: Learn Python

Time: 4 to 6 weeks | Python first

Before AI. Before ML. Before anything. You need one skill: Python. Not because it's the only language, but because the entire AI world runs on it. Every framework, tool, and tutorial is Python first. Python is beginner-friendly: go from zero to writing real code in a few weeks.

WHAT TO LEARN

- → Variables, data types, ops
- → Loops & conditionals
- → Functions & modules
- → Lists, dicts, tuples
- → File handling, basic OOP
- → NumPy, Pandas (basics)

PRACTICE PLATFORMS

- HackerRank track
- LeetCode (Easy)
- Exercism.io exercises

FREE RESOURCES

- Python for Everybody
- CS50P — Harvard
- freeCodeCamp (4h)
- Tech With Tim
- Corey Schafer
- Automate Boring Stuff
- W3Schools Python

Milestone: Write a Python script that reads a CSV file, cleans the data, and prints basic statistics.

STEP 2: The Math You Need



Time: 4 to 6 weeks (parallel with Python)

Everyone says "you need math for AI." That's true, but you don't need a PhD. You need enough math to understand what is happening inside the models. Learn math alongside ML — context makes it stick 10x faster.

1. LINEAR ALGEBRA
Vectors, matrices, dot products, transposes, eigenvalues.

2. STATS & PROB
Mean, median, variance, distributions, Bayes theorem, correlation.

3. CALCULUS
Derivatives & gradients. Understand the slope concept.

FREE RESOURCES

- Khan Academy Math
- 3Blue1Brown (LA)
- 3Blue1Brown (Calculus)
- StatQuest with Josh
- Math for ML (Imperial)
- Gilbert Strang (MIT OCW)
- Neural Networks (3B1B)

Tip: Don't try to master math before starting ML. Learn math alongside it. Context makes it stick 10x faster.

STEP 3: Understand Machine Learning

Time: 6 to 8 weeks | Library: Scikit-learn

Machine Learning is teaching computers to learn from data instead of being explicitly programmed. You already understand the concept. Now let's make it technical.

CORE CONCEPTS

- → Supervised / Unsupervised / RL
- → Train, test, validation splits
- → Overfitting & underfitting
- → Regression & Classification
- → Clustering & Decision Trees
- → Accuracy, Precision, Recall, F1
- → Cross validation & Feature Eng.



FREE RESOURCES

- Google ML Crash Course
- Andrew Ng ML Course
- Kaggle Intro to ML
- Kaggle Intermediate ML
- StatQuest ML Playlist
- sentdex ML with Python
- Codebasics ML Tutorial
- Hands-On ML (Geron)

STEP 4: Data is Everything



Time: 3 to 4 weeks | 80% of real AI work

Here's what nobody tells beginners: 80% of real AI work is cleaning, preparing, and understanding data. Garbage in, garbage out. The best algorithm fails on bad data. This step is unglamorous but vital.

WHAT TO LEARN

- → EDA & data collection
- → Handling missing values
- → Removing outliers & duplicates
- → Scaling & normalization
- → Categorical encoding
- → Data visualization & eng.

LIBRARIES & DATASETS

- Pandas
- NumPy
- Matplotlib
- Seaborn

DATASETS SEARCH

- Kaggle Datasets
- UCI ML Repo
- Google Dataset Search

FREE RESOURCES

- Kaggle Pandas
- Kaggle Data Vis
- Keith Galli Pandas
- Corey Schafer Pandas
- freeCodeCamp Data Analysis
- Alex the Analyst

Milestone: Take a messy real-world dataset, clean it completely, visualize 5 key insights, and write your findings.

STEP 5: Build First Projects



Build now. Don't wait. | Learning by building

Reading about AI is not learning AI. Watching tutorials is not learning AI. Building is learning. After Steps 1 to 4, you have enough to build real things. Start now. Don't wait for Step 10.

BEGINNER PROJECTS TO BUILD

- → House price predictor (regression)
- → Spam email classifier (classification)
- → Customer churn predictor
- → Iris flower classifier (classic ML)
- → Movie recommendation system
- → Titanic survival prediction (Kaggle)
- → Stock price visualizer (data + vis)
- → Sentiment analyzer on tweets (NLP)

TOOLS & PLATFORMS

- Scikit-learn
- Pandas
- Jupyter
- Google Colab

FREE PLATFORMS

- Kaggle Notebooks
- Google Colab (Free GPU)
- GitHub (Code sharing)
- Hugging Face Spaces

Pro Tip: Every project you build — write a LinkedIn post about it. What you built, what you learned, what was hard. This alone will get you noticed faster than any resume. (Ongoing from Step 3 onwards).

STEP 6: Deep Learning



Time: 8 to 12 weeks | PyTorch / TensorFlow

You've learned how machines learn from data. Now learn how machines think in layers. Deep Learning is the engine behind ChatGPT, Midjourney, Face ID, and self-driving cars.

CORE CONCEPTS

- → What is a Neural Network? Layers
- → ReLU, Sigmoid, Softmax activations
- → Forward pass & backpropagation
- → CNNs (images) & RNNs (sequences)
- → Transformers (GPT, BERT, modern)
- → Transfer learning & Regularization

FRAMEWORKS

PYTORCH
Industry standard, preferred for research & production.

TENSORFLOW
Widely used, slightly more beginner friendly.

FREE RESOURCES

- Andrew Ng DL Spec
- fast.ai Deep Learning
- Karpathy Zero to Hero
- PyTorch Official Tutorials
- Daniel Bourke PyTorch
- Sentdex NN from Scratch
- Lex Fridman DL MIT
- freeCodeCamp PyTorch

STEP 7: GenAI & LLMs

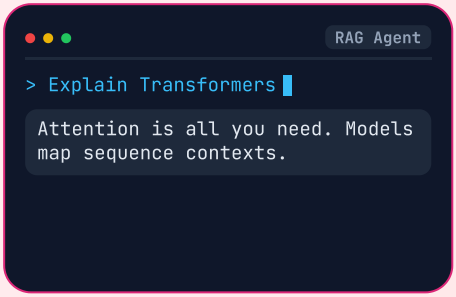


Time: 6 to 8 weeks

Entering AI in 2025+

CORE CONCEPTS

- → What is an LLM? (Transformers)
- → Prompt Engineering & fine-tuning
- → RAG (Retrieval Augmented Gen)
- → Vector databases (Pinecone, etc.)
- → AI Agents taking actions
- → LangChain, LangGraph frameworks
- → Embeddings (words to numbers)



FREE COURSES & GUIDES

- Prompt Eng. for Devs
- LangChain LLM Dev
- Building ChatGPT APIs
- Hugging Face NLP Course
- Karpathy Let's build GPT
- Sam Witteveen Tutorials
- James Briggs Vector DB
- DeepLearning.AI Short Courses

OpenAI API · Hugging Face · LangChain · Pinecone · Gradio/Streamlit

STEP 8: Pick Your Lane

AI is not one career. It's many. By now you have a strong foundation. The next step is choosing where you want to go deep. Don't try to master everything. Pick the lane that excites you most and specialize.

THE LANES

- 🗣️ **NLP Engineer**
  LLMs, bots, text
- 👀 **Computer Vision**
  Images, objects, video
- 🎨 **Gen AI Engineer**
  LLMs, diffusions, tools
- 📊 **Data Scientist**
  Stats, business insights
- 🧠 **ML Engineer**
  Training, optimization
- ⚙️ **MLOps Engineer**
  Pipelines, infra, ops
- 🤖 **AI Engineer**
  AI products & APIs
- 🧭 **AI Product Manager**
  Strategy & roadmaps

ℹ️ **Not sure which one?** Go back to the AI Careers carousel we posted earlier. Every role is mapped out with their tech stacks, tools, and responsibilities. Your interest is your direction.

STEP 9: Build a Portfolio



> Nobody gets hired for knowing AI. They get hired for showing AI. A portfolio is your proof of work. It speaks louder than any certificate. (Build continuously alongside every step).

PORTFOLIO NEEDS

- ✓ 3 to 5 projects min
- ✓ Clean code & README
- ✓ Deployed live project
- ✓ Lane specific project
- ✓ Explain what & why

SHOWCASE & DEPLOYS

- GitHub Profile
- LinkedIn Posts
- Personal Website
- Kaggle Profile

PROJECT IDEAS & DEPLOYS

PROJECTS BY LEVEL

Beg: Price prediction, Spam classifier, Image classifier
Int: Sentiment app, Object detection, Movie recommender
Adv: RAG chatbot on docs, Fine-tuned LLM, AI Agent

DEPLOYS TOOLS

Hugging Face Spaces, Streamlit Cloud, Gradio, Render, Railway, GitHub.

Milestone: Have at least one live working AI app someone can open in their browser and use right now.

STEP 10: Learn in Public

The loneliest thing about learning AI is doing it alone. Most people quit not because the content is too hard, but because they have no one to learn with. Document your journey — people follow journeys.

LEARN IN PUBLIC

- → Post what you learned on LinkedIn
- → Share imperfect projects
- → Ask questions without shame
- → Write simple explanations

STAY UPDATED

- → The Batch — DeepLearning.AI
- → Import AI — Jack Clark
- → Hugging Face Blog latest
- → Arxiv Sanity simplified papers

FREE COMMUNITIES TO JOIN

- ResnurAI concepts & news 🌱
- Hugging Face Discord
- Kaggle Forums
- r/MachineLearning
- r/learnmachinelearning
- AI/ML Discord servers
- Papers With Code

> One LinkedIn post per week about what you're learning will do more for your career than 10 certificates. The more you share, the faster you grow.

STEP 11: First Opportunity



You don't need to wait until you're "ready" to apply. You need a portfolio, a few projects, and the confidence to start.

WAYS TO GET OPPORTUNITIES

- → Kaggle Competitions: compete & get noticed
- → Open Source: HuggingFace, Scikit-learn, LangChain
- → Internships: apply early, portfolio matters
- → Freelance Projects: Upwork, Freelancer, LinkedIn
- → Research: email professors, apply as assistant
- → Hackathons: DevPost, MLH, Unstop (visibility)
- → LinkedIn: optimize profile, post, connect

JOB PLATFORMS

- LinkedIn Jobs
- Wellfound Startup
- Kaggle Jobs Board
- Hugging Face Jobs
- AI Jobs (aijobs.net)
- RemoteAI.jobs

> **Tip:** Your first role doesn't have to be your dream role. It has to get you in the room. From there, everything accelerates.

The Complete Roadmap Glance



- **Step 1 → Learn Python** — 4-6 Weeks
- **Step 2 → Math Foundations (Linear Alg, Stats, Calc)** — 4-6 Weeks
- **Step 3 → Machine Learning Fundamentals** — 6-8 Weeks
- **Step 4 → Data Manipulation & Analysis** — 3-4 Weeks
- **Step 5 → Build Your First Real Projects** — Ongoing
- **Step 6 → Deep Learning (Neural Nets, Transformers)** — 8-12 Weeks
- **Step 7 → Generative AI & LLMs (RAG, Agents)** — 6-8 Weeks
- **Steps 8 - 11 → Pick Lane, Portfolio, Public, Jobs** — Continuous

1 - 2 HOURS DAILY	3 - 4 HOURS DAILY
10 to 14 Months	6 to 8 Months



A year from now you will wish you had started today.

Every expert you admire was once exactly where you are: confused, overwhelmed, and unsure where to begin. The difference? They started anyway. You have the roadmap, the resources, and the community. The only thing left is slide 4. Start there. Today.

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